Numerical verification of the mean and variance asymptotics

This note collects the numerical checks of the asymptotic means and variances of the number of cliques $\mathbf{C}_n$, the number of edges $\mathbf{M}_n$ and the degree $\mathbf{D}_n$ of a uniformly chosen vertex of $\mathbf{CG}_{n,p}$. We consider the critical regime $p = 1/2$ and the subcritical regime $p = 1/4$. All tables are produced by `python verification.py`.

For each regime, we compare

1. the **exact** moments with the **saddle-point** moments for $n \le 1000$, and
2. the **saddle-point** moments with the **asymptotes** from the paper for n up to 10^{256}.

The second step is needed because the asymptotes converge slowly: in the subcritical regime, their relative corrections are of order $1/\tau_n = O(1/\sqrt{\log n})$. The saddle-point moments can be evaluated for any n, whereas the exact moments can only be computed for moderate n.

Throughout, $w = p/(1-p)$, $a = \log(1/w)$ and W denotes the Lambert W-function.

Exact moments

By Corollary 2.5, the size $S_n = \mathbf{D}_n + 1$ of the clique containing a given vertex satisfies

$$\mathbb{P}(S_n = s) = \binom{n-1}{s-1} w^{\binom{s}{2}} \frac{B_{n-s}(w)}{B_n(w)}, \qquad s = 1, \dots, n, \tag{2}$$

where $B_n(w)$ is computed with the recursion of Corollary 2.6. Given $S_n = s$, the remaining $n - s$ vertices form a $\mathbf{CG}_{n-s,p}$, so that

$$\mathbf{C}_n \overset{d}{=} 1 + \mathbf{C}_{n-S_n}, \qquad \mathbf{M}_n \overset{d}{=} \binom{S_n}{2} + \mathbf{M}_{n-S_n}, \qquad \mathbf{D}_n = S_n - 1, \tag{3}$$

where, conditionally on $S_n = s$, $\mathbf{C}_{n-s}$ and $\mathbf{M}_{n-s}$ are the statistics of a $\mathbf{CG}_{n-s,p}$ on the remaining vertices. The means follow by conditioning on S_n, and the variances follow from the law of total variance. Everything is computed in arbitrary precision (mpmath) for all $n \le 1000$.

Saddle-point moments

Let

$$C_r(w, x) = \sum_{j \ge 1} j^r \, w^{\binom{j}{2}} \frac{x^j}{j!}, \tag{4}$$

and let x_n be the saddle point, i.e. the solution of $C_1(w, x_n) = n$. Writing $C_r = C_r(w, x_n)$, the saddle-point moments are

$$\begin{aligned} \mu_{\mathbf{C}} &= C_0, & \sigma^2_{\mathbf{C}} &= C_0 - \frac{C_1^2}{C_2}, \\ \mu_{\mathbf{M}} &= \frac{C_2 - C_1}{2}, & \sigma^2_{\mathbf{M}} &= \frac{C_2 C_4 - C_3^2}{4 C_2}, \\ \mu_{\mathbf{D}} &= \frac{C_2}{C_1} - 1, & \sigma^2_{\mathbf{D}} &= \frac{C_3}{C_1} - \left(\frac{C_2}{C_1}\right)^2. \end{aligned} \tag{5}$$

For $\mathbf{C}_n$ and $\mathbf{M}_n$, these are the quantities $\mu_{n,p} = H'(0)$ and $\sigma^2_{n,p} = H''(0)$ from the proofs of Propositions 5.2 and 5.4 (see (5.13), (5.14) and (5.22), (5.23)). The expressions for $\mathbf{D}_n$ are given in Proposition 5.6.

The saddle point is parametrised as $x_n = \tau_n w^{1/2-\tau_n}$, so that τ_n is the solution of $C_1\left(w, \tau_n w^{1/2-\tau_n}\right) = n$ (Lemma B.3).

For $p = 1/2$ we have $w = 1$, $C_1(1, x) = xe^x$ and hence $x_n = \tau_n = W(n)$. Writing $W = W(n)$, the saddle-point moments then evaluate to

$$\begin{aligned} \mu_{\mathbf{C}} &= \frac{n}{W} - 1, & \sigma^2_{\mathbf{C}} &= \frac{n}{W(1+W)} - 1, \\ \mu_{\mathbf{M}} &= \frac{nW}{2}, & \sigma^2_{\mathbf{M}} &= \frac{nW(2+2W+W^2)}{4(1+W)}, \\ \mu_{\mathbf{D}} &= W, & \sigma^2_{\mathbf{D}} &= W. \end{aligned} \tag{6}$$

In particular, $\mu_{\mathbf{M}}$, $\mu_{\mathbf{D}}$ and $\sigma^2_{\mathbf{D}}$ coincide exactly with their asymptotes below, so the corresponding tables only show rounding errors.

Asymptotes

Critical regime $p = 1/2$ (Theorems 1.1 (ii), 1.3 (ii) and 1.4 (ii); Lemmas 4.1 and 4.8):

$$\begin{aligned} \mathbb{E}[\mathbf{C}_n] &\sim \frac{n}{W(n)}, & \mathrm{Var}(\mathbf{C}_n) &\sim \frac{n}{W(n)^2}, \\ \mathbb{E}[\mathbf{M}_n] &\sim \frac{nW(n)}{2}, & \mathrm{Var}(\mathbf{M}_n) &\sim \frac{nW(n)^2}{4}, \\ \mathbb{E}[\mathbf{D}_n] &\sim W(n), & \mathrm{Var}(\mathbf{D}_n) &\sim W(n). \end{aligned} \tag{7}$$

The last asymptote follows from the Poisson limit of $\mathbf{D}_n$ with matching moments.

Subcritical regime $p < 1/2$ (Propositions 5.2, 5.4 and 5.6):

$$\begin{aligned} \mathbb{E}[\mathbf{C}_n] &\sim \frac{n}{\tau_n}, & \mathrm{Var}(\mathbf{C}_n) &\sim \frac{n}{\tau_n^3}\mathrm{Var}(T), \\ \mathbb{E}[\mathbf{M}_n] &\sim \frac{n\tau_n}{2}, & \mathrm{Var}(\mathbf{M}_n) &\sim \frac{n\tau_n}{4}\mathrm{Var}(T), \\ \mathbb{E}[\mathbf{D}_n] &= \tau_n - 1 + \frac{e_{w,1}(\tau_n)}{e_{w,0}(\tau_n)} + o(1), & \mathrm{Var}(\mathbf{D}_n) &= \mathrm{Var}(T) + o(1). \end{aligned} \tag{8}$$

Here $e_{w,\ell}(\tau) = \sum_{t+\tau\in\mathbb{Z}} t^\ell w^{t^2/2}$ are the periodic functions of Lemma B.2, and

$$\mathrm{Var}(T) = \frac{e_{w,0}(\tau_n)e_{w,2}(\tau_n) - e_{w,1}(\tau_n)^2}{e_{w,0}(\tau_n)^2} \tag{9}$$

is the periodic factor of Remark 5.3.

Asymptotics of τ_n (Lemma B.3):

$$\tau_n = \sqrt{\frac{2\log n}{a}} - \frac{1}{a} + O\left(\frac{\log\log n}{\sqrt{\log n}}\right). \tag{10}$$

Critical regime $p = 1/2$

Exact vs saddle point

$\mathbb{E}[\mathbf{C}_n]$

n	exact	saddle point	exact / saddle point -1
50	16.5744	16.4771	+5.904e-03
100	28.6253	28.5366	+3.108e-03
250	59.9296	59.8506	+1.320e-03
500	106.074	106.001	+6.862e-04
1000	189.558	189.491	+3.550e-04

$\mathrm{Var}(\mathbf{C}_n)$

n	exact	saddle point	exact / saddle point -1
50	3.53863	3.5267	+3.383e-03
100	5.74575	5.73486	+1.900e-03
250	10.9212	10.9118	+8.586e-04
500	17.8703	17.862	+4.637e-04
1000	29.4877	29.4804	+2.480e-04

$\mathbb{E}[\mathbf{M}_n]$

n	exact	saddle point	exact / saddle point -1
50	70.7474	71.5223	-1.083e-02
100	168.384	169.282	-5.305e-03
250	512.482	513.553	-2.085e-03
500	1167	1168.21	-1.033e-03
1000	2623.45	2624.8	-5.134e-04

$\mathrm{Var}(\mathbf{M}_n)$

n	exact	saddle point	exact / saddle point -1
50	142.978	147.332	-2.955e-02
100	384.729	390.503	-1.479e-02
250	1353.92	1361.99	-5.925e-03
500	3406.36	3416.5	-2.967e-03
1000	8399.48	8411.98	-1.486e-03

$\mathbb{E}[\mathbf{D}_n]$

n	exact	saddle point	exact / saddle point -1
50	2.8299	2.86089	-1.083e-02
100	3.36767	3.38563	-5.305e-03
250	4.09986	4.10842	-2.085e-03
500	4.66801	4.67284	-1.033e-03
1000	5.24691	5.2496	-5.134e-04

$\mathrm{Var}(\mathbf{D}_n)$

n	exact	saddle point	exact / saddle point -1
50	2.78617	2.86089	-2.612e-02
100	3.34106	3.38563	-1.316e-02
250	4.08653	4.10842	-5.329e-03
500	4.66028	4.67284	-2.689e-03
1000	5.24249	5.2496	-1.356e-03

Saddle point vs asymptote

$\mathbb{E}[\mathbf{C}_n]$ **vs** $n/W(n)$

n	saddle point	asymptote	saddle point / asymptote -1
1000	189.491	190.491	-5.250e-03
10000	1381.77	1382.77	-7.232e-04
10^8	6.38203e+06	6.38203e+06	-1.567e-07
10^{16}	2.99987e+14	2.99987e+14	-3.775e-15
10^{32}	1.44005e+30	1.44005e+30	-1.554e-15
10^{64}	7.02214e+61	7.02214e+61	+1.110e-15
10^{128}	3.45944e+125	3.45944e+125	-4.441e-16
10^{256}	1.71499e+253	1.71499e+253	-2.220e-16

$\mathrm{Var}(\mathbf{C}_n)$ **vs** $n/W(n)^2$

n	saddle point	asymptote	saddle point / asymptote -1
1000	29.4804	36.2867	-1.876e-01
10000	166.978	191.206	-1.267e-01
10^8	382867	407303	-5.999e-02
10^{16}	8.73713e+12	8.99923e+12	-2.913e-02
10^{32}	2.04429e+28	2.07373e+28	-1.420e-02
10^{64}	4.89666e+59	4.93104e+59	-6.973e-03
10^{128}	1.19265e+123	1.19677e+123	-3.448e-03
10^{256}	2.93616e+250	2.94119e+250	-1.712e-03

$\mathbb{E}[\mathbf{M}_n]$ **vs** $nW(n)/2$

n	saddle point	asymptote	saddle point / asymptote -1
1000	2624.8	2624.8	-2.220e-16
10000	36159.2	36159.2	+0.000e+00
10^8	7.8345e+08	7.8345e+08	+2.220e-16
10^{16}	1.66674e+17	1.66674e+17	+4.441e-16
10^{32}	3.47211e+33	3.47211e+33	+1.554e-15
10^{64}	7.12034e+65	7.12034e+65	-1.332e-15
10^{128}	1.44532e+130	1.44532e+130	+4.441e-16
10^{256}	2.91547e+258	2.91547e+258	+2.220e-16

$\mathrm{Var}(\mathbf{M}_n)$ **vs** $nW(n)^2/4$

n	saddle point	asymptote	saddle point / asymptote -1
1000	8411.98	6889.58	+2.210e-01
10000	151025	130749	+1.551e-01
10^8	6.55316e+09	6.13794e+09	+6.765e-02
10^{16}	2.86378e+18	2.77802e+18	+3.087e-02
10^{32}	1.22316e+35	1.20556e+35	+1.460e-02
10^{64}	5.10577e+67	5.06992e+67	+7.071e-03
10^{128}	2.0962e+132	2.08895e+132	+3.471e-03
10^{256}	8.51455e+260	8.49995e+260	+1.718e-03

$\mathbb{E}[\mathbf{D}_n]$ **vs** $W(n)$

n	saddle point	asymptote	saddle point / asymptote -1
1000	5.2496	5.2496	+0.000e+00
10000	7.23185	7.23185	+0.000e+00
10^8	15.669	15.669	+2.220e-16
10^{16}	33.3348	33.3348	+4.441e-16
10^{32}	69.4422	69.4422	+1.554e-15
10^{64}	142.407	142.407	-9.992e-16
10^{128}	289.064	289.064	+4.441e-16
10^{256}	583.093	583.093	+4.441e-16

$\mathrm{Var}(\mathbf{D}_n)$ **vs** $W(n)$

n	saddle point	asymptote	saddle point / asymptote -1
1000	5.2496	5.2496	+0.000e+00
10000	7.23185	7.23185	+2.220e-16
10^8	15.669	15.669	+4.441e-16
10^{16}	33.3348	33.3348	-3.664e-15
10^{32}	69.4422	69.4422	+5.329e-15
10^{64}	142.407	142.407	+7.772e-15
10^{128}	289.064	289.064	+1.532e-14
10^{256}	583.093	583.093	+2.864e-14

Subcritical regime $p = 1/4$

Exact vs saddle point

$\mathbb{E}[\mathbf{C}_n]$

n	exact	saddle point	exact / saddle point -1
50	26.9574	26.898	+2.209e-03
100	49.2728	49.2194	+1.085e-03
250	110.639	110.593	+4.230e-04
500	205.73	205.688	+2.072e-04
1000	385.055	385.016	+1.014e-04

$\mathrm{Var}(\mathbf{C}_n)$

n	exact	saddle point	exact / saddle point -1
50	3.58817	3.57924	+2.494e-03
100	6.07578	6.06796	+1.288e-03
250	12.1397	12.1333	+5.296e-04
500	20.5784	20.5728	+2.690e-04
1000	35.1249	35.1201	+1.360e-04

$\mathbb{E}[\mathbf{M}_n]$

n	exact	saddle point	exact / saddle point -1
50	28.4726	28.6049	-4.623e-03
100	65.7334	65.8711	-2.091e-03
250	192.245	192.39	-7.495e-04
500	425.108	425.256	-3.492e-04
1000	928.845	928.997	-1.640e-04

$\mathrm{Var}(\mathbf{M}_n)$

n	exact	saddle point	exact / saddle point -1
50	13.9719	14.0644	-6.580e-03
100	31.6817	31.7799	-3.089e-03
250	91.2613	91.3662	-1.148e-03
500	200.233	200.343	-5.481e-04
1000	435.006	435.121	-2.628e-04

$\mathbb{E}[\mathbf{D}_n]$

n	exact	saddle point	exact / saddle point -1
50	1.1389	1.14419	-4.623e-03
100	1.31467	1.31742	-2.091e-03
250	1.53796	1.53912	-7.495e-04
500	1.70043	1.70103	-3.492e-04
1000	1.85769	1.85799	-1.640e-04

$\mathrm{Var}(\mathbf{D}_n)$

n	exact	saddle point	exact / saddle point -1
50	0.539039	0.542534	-6.441e-03
100	0.569778	0.571522	-3.051e-03
250	0.601588	0.602275	-1.141e-03
500	0.620865	0.621203	-5.444e-04
1000	0.637103	0.63727	-2.610e-04

Saddle point vs asymptote

$\mathbb{E}[\mathbf{C}_n]$ **vs** n/τ_n

n	saddle point	asymptote	saddle point / asymptote -1
1000	385.016	372.271	+3.423e-02
10000	3208.56	3122.71	+2.749e-02
10^8	2.08943e+07	2.0609e+07	+1.385e-02
10^{16}	1.39201e+15	1.38253e+15	+6.862e-03
10^{32}	9.44653e+30	9.41444e+30	+3.408e-03
10^{64}	6.49499e+62	6.48398e+62	+1.698e-03
10^{128}	4.50567e+126	4.50186e+126	+8.474e-04
10^{256}	3.14457e+254	3.14324e+254	+4.234e-04

$\mathrm{Var}(\mathbf{C}_n)$ **vs** $n\tau_n^{-3}\,\mathrm{Var}(T)$

n	saddle point	asymptote	saddle point / asymptote -1
1000	35.1201	46.9607	-2.521e-01
10000	220.626	277.172	-2.040e-01
10^8	682761	796753	-1.431e-01
10^{16}	2.15836e+13	2.40534e+13	-1.027e-01
10^{32}	7.03493e+28	7.59521e+28	-7.377e-02
10^{64}	2.34998e+60	2.48131e+60	-5.293e-02
10^{128}	7.99022e+123	8.30482e+123	-3.788e-02
10^{256}	2.7503e+251	2.82675e+251	-2.704e-02

$\mathbb{E}[\mathbf{M}_n]$ **vs** $n\tau_n/2$

n	saddle point	asymptote	saddle point / asymptote -1
1000	928.997	1343.11	-3.083e-01
10000	11734	16011.7	-2.672e-01
10^8	1.97383e+08	2.42613e+08	-1.864e-01
10^{16}	3.14849e+16	3.61657e+16	-1.294e-01
10^{32}	4.83266e+32	5.31099e+32	-9.006e-02
10^{64}	7.2262e+64	7.71131e+64	-6.291e-02
10^{128}	1.06168e+129	1.11065e+129	-4.409e-02
10^{256}	1.54144e+257	1.59072e+257	-3.098e-02

$\mathrm{Var}(\mathbf{M}_n)$ **vs** $n\tau_n \,\mathrm{Var}(T)/4$

n	saddle point	asymptote	saddle point / asymptote -1
1000	435.121	611.274	-2.882e-01
10000	5440	7287.26	-2.535e-01
10^8	9.06667e+07	1.10418e+08	-1.789e-01
10^{16}	1.44269e+16	1.64597e+16	-1.235e-01
10^{32}	2.21134e+32	2.41714e+32	-8.514e-02
10^{64}	3.30274e+64	3.50957e+64	-5.893e-02
10^{128}	4.84756e+128	5.0548e+128	-4.100e-02
10^{256}	7.0323e+256	7.23966e+256	-2.864e-02

$\mathbb{E}[\mathbf{D}_n]$ **vs** $\tau_n - 1 + e_{w,1}/e_{w,0}$

n	saddle point	asymptote	saddle point / asymptote -1
1000	1.85799	1.68621	+1.019e-01
10000	2.34679	2.20235	+6.559e-02
10^8	3.94766	3.85226	+2.477e-02
10^{16}	6.29698	6.23313	+1.024e-02
10^{32}	9.66532	9.62198	+4.505e-03
10^{64}	14.4524	14.4226	+2.064e-03
10^{128}	21.2337	21.2131	+9.721e-04
10^{256}	30.8287	30.8143	+4.665e-04

$\mathrm{Var}(\mathbf{D}_n)$ **vs** $\mathrm{Var}(T)$

n	saddle point	asymptote	saddle point / asymptote -1
1000	0.63727	0.91024	-2.999e-01
10000	0.677085	0.910239	-2.561e-01
10^8	0.751328	0.910239	-1.746e-01
10^{16}	0.801304	0.910239	-1.197e-01
10^{32}	0.834948	0.91024	-8.272e-02
10^{64}	0.857843	0.91024	-5.756e-02
10^{128}	0.873591	0.910239	-4.026e-02
10^{256}	0.884519	0.910239	-2.826e-02

Asymptotics of τ_n for $p = 1/4$

n	τ_n	$\sqrt{2 \log n/a} - 1/a$	difference
1000	2.68621	2.63594	+5.027e-02
10000	3.20235	3.18454	+1.781e-02
10^8	4.85226	4.88065	-2.840e-02
10^{16}	7.23313	7.27932	-4.619e-02
10^{32}	10.622	10.6715	-4.957e-02
10^{64}	15.4226	15.4689	-4.626e-02
10^{128}	22.2131	22.2533	-4.028e-02
10^{256}	31.8143	31.848	-3.367e-02